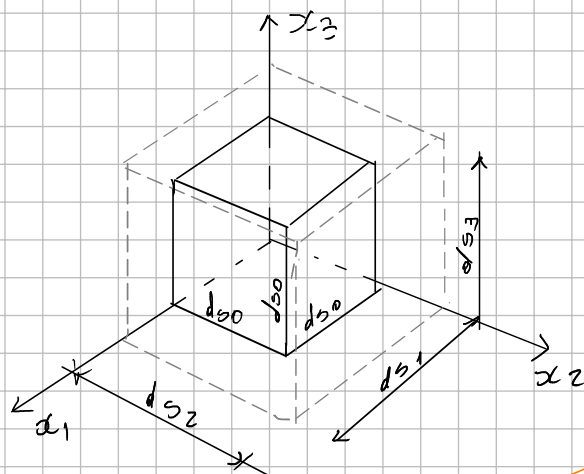




$$\frac{ds - ds_0}{ds_0} = \underline{n} \cdot \underline{E} \cdot \underline{n} \quad \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} E_{xx} & E_{xy} & E_{yx} \\ E_{xy} & E_{yy} & E_{yz} \\ E_{yx} & E_{yz} & E_{zz} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\underline{n} \rightarrow \text{dirección } x \Rightarrow \frac{ds_1 - ds_0}{ds_0} = E_{xx} = \epsilon_{11} \quad \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} E_{xx} \\ E_{xy} \\ E_{yz} \end{bmatrix} = E_{xx}$$

$$\underline{n} \rightarrow \text{dirección } y \Rightarrow \frac{ds_2 - ds_0}{ds_0} = E_{yy} = \epsilon_{22}$$

$$\underline{n} \rightarrow \text{dirección } z \Rightarrow \frac{ds_3 - ds_0}{ds_0} = E_{zz} = \epsilon_{33}$$

$$E_{xx} = \frac{\partial u_x}{\partial x} \quad E_{yy} = \frac{\partial u_y}{\partial y} \quad E_{zz} = \frac{\partial u_z}{\partial z}$$

$$ds_1 = ds_0 (1 + \epsilon_{11})$$

$$ds_2 = ds_0 (1 + \epsilon_{22})$$

$$ds_3 = ds_0 (1 + \epsilon_{33})$$

$$V = ds_1 \cdot ds_2 \cdot ds_3$$

$$V_0 = ds_0 \cdot ds_0 \cdot ds_0$$

$$V = \underbrace{ds_0^3}_{V_0} \cdot (1 + \epsilon_{11}) \cdot (1 + \epsilon_{22}) \cdot (1 + \epsilon_{33})$$

$$\begin{aligned} \chi &= \frac{\Delta V}{V_0} = \frac{V - V_0}{V_0} = \frac{[(1 + \epsilon_{11}) \cdot (1 + \epsilon_{22}) \cdot (1 + \epsilon_{33}) - 1] V_0}{V_0} \\ &= 1 + \epsilon_{33} + \epsilon_{22} + \epsilon_{11} + \underbrace{\epsilon_{11} \cdot \epsilon_{33}}_{\approx 0} + \underbrace{\epsilon_{22} \cdot \epsilon_{33}}_{\approx 0} + \underbrace{\epsilon_{11} \cdot \epsilon_{22}}_{\approx 0} + \underbrace{\epsilon_{11} \cdot \epsilon_{22} \cdot \epsilon_{33}}_{\approx 0} - 1 \\ &= \epsilon_{11} + \epsilon_{22} + \epsilon_{33} = I_1(\underline{\epsilon}) \end{aligned}$$

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